Seasonality in the Cross-Section of Expected Stock Returns†

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Abstract

This paper introduces seasonality into a model of expected stock returns. We confirm previous findings that there is no evidence for cross-sectional variation in expected stock returns when we restrict the means to be constant throughout the year. Yet, we show there is substantial variation when considering each month of the year separately. Applying a seasonal structure we estimate an annualized standard deviation of 13.8%. There is strong evidence stocks have distinct expected returns in January, February, …., December. The estimated seasonal variation in expected returns is positive in every calendar month and especially high during October, December, and January. This structure is independent of industry, size, and earnings announcements. These results support the inclusion of seasonal structure into asset-pricing models.

Introduction

This paper introduces seasonality into a model of expected stock returns. The traditional asset-pricing literature focuses on cross-sectional differences in unconditional expected returns across stocks (e.g., Fama and MacBeth (1973), Fama and French (1992)). The modern literature extends this to differences in conditional expected returns (e.g., Jaganathan and Wang (1996), Lettau and Ludvigson (2001), and Chordia and Shivakumar (2002)). These papers use various conditioning variables, such as labor income, credit spread, and term spread, to predict expected returns. In this paper we use seasonal dummy variables as instruments for conditioning information.

Seasonal variables provide natural conditioning information for tests of dynamic asset-pricing models. Seasonal economic influences on expected returns may originate in several ways. It is important to note that annual cycles can emerge endogenously in economic models and are consistent with rational economic equilibrium (Huberman (1988)). Seasonal variation in expected return may be a rational equilibrium response to exogenous influences. In this vein Ogden (2003) has developed a general seasonal model for stocks and bonds. There are also behavioral models that characterize the seasonality of stocks returns in terms of weather and vacation patterns (Bouman and Jacobsen (2002), Kamstra, Kramer, and Levi (2003)).

Despite extensive application of seasonality to market index returns (e.g., the January effect documented by Rozeff and Kinney (1976)) there have been relatively few papers that use seasonality to explain the cross-sectional *differences* in expected returns among stocks. Keim (1983) and Reinganum (1983) find that small stocks outperformed large stocks in January, and Tinic and West (1984) find that high-beta stocks outperform low-beta stocks in January. However, the empirical literature has not fully explored general seasonal variation across individual stocks.

The issue of seasonality has large consequences for interpreting asset-pricing models. Empirical studies of models such as the CAPM or Fama and French's (1993) three-factor model have found only modest differences in expected returns among common stocks. Indeed, direct tests by Jegadeesh and Titman (2002) found no evidence of cross-sectional variation in expected returns. But when we apply their methodology with seasonal structure we estimate an annualized standard deviation of 13.8%. In other words the conclusion changes from negligible variation in expected returns to extremely large variation in expected returns.

This paper pursues a simple approach to detect seasonal variation in the cross-section of expected stock returns. It is based on the approach of Conrad and Kaul (1998) and Jegadeesh and Titman (2002) to measuring the cross-sectional variance in expected return across individual stocks. These studies note that stocks with high historical returns tend to have high expected returns. Consequently a strategy that buys stocks with high historical returns and sells stocks with low historical returns should earn high expected returns in future months. This intuitively promising approach has produced disappointing results by finding little robust evidence of variation in expected returns. We confirm this approach provides no evidence of return variation when using all months. But we show new evidence of substantial variation within each month of the year. Stocks with high historical returns in a particular calendar month tend to have high future returns in that same calendar month. In other words some stocks earn high expected returns every

January, others in February, March, etc. The cross-sectional variance of expected returns appears particularly large in January, October, and December.

A simple approach to exploiting this effect and measuring economic significance is to form winner-loser decile spreads based on seasonal returns in previous years. These strategies produce significantly positive returns for up to 20 years. The strategies are unaffected by demeaning stock returns across all months, but their profits vanish when returns are demeaned seasonally. The profitability of seasonal strategies is not associated with industry, size, or earnings announcements. These results support the inclusion of seasonal structure into asset pricing models.

The rest of this paper is organized as follows. Section 1 introduces a simple model of stock returns and generalizes it to allow seasonal variation. Section 2 uses decile spreads to confirm the existence of a seasonal effect in stock returns. Section 3 performs sensitivity analysis with industry and size. Section 4 considers earnings announcements. A discussion of aspects is included in Section 5, and Section 6 concludes.

1. The Cross-Section of Expected Stock Returns

This section begins by analyzing a stationary model of stock returns. It then extends the model to allow seasonality in expected returns. The seasonal version specifically allows expected returns to vary separately across different calendar months instead of being constant throughout the whole year. In this manner we can easily modify the methodology to reach different conclusion. In particular, applying the same methodology to separate calendar months instead of combined months produces dramatically different conclusions.

1.1 A Stationary Model of Stock Returns

We begin with the one-factor model used by Jegadeesh and Titman (1993) and Conrad and Kaul (1998). This model describes the monthly returns on a stock, r_{it}, in excess of the equally weighted market index return $\bar{r}_t$

$$r_{it} - \bar{r}_t = \mu_i + b_i f_t + e_{it} \tag{1}$$

$$E[f_t] = E[e_t] = 0,$$

$$Cov(f_t, e_{it}) = 0,$$

where μ_i is the unconditional expected return on stock i in month t in excess of the market index, b_i is the factor sensitivity of stock i, f_t is the (unconditional) unexpected return on a factor-mimicking portfolio, and e_{it} is the firm-specific component of return. We use a monthly time interval instead of a longer interval in order to generalize the model by allowing expected returns to vary within the year.

Now following Lo and MacKinlay (1990) and Jegadeesh and Titman (1993) we investigate the implications of the model for returns on weighted relative strength strategies. Lehman (1990) pioneered the use of these strategies for short-horizon returns; we extend this methodology to more distant returns. A weighted relative strength strategy with lag k is a portfolio that buys or sells stocks at time t using weights $w_{it}(k)$ based on the lag-k historical return relative to the market

$$w_{it}(k) = \frac{1}{N}(r_{it-k} - \bar{r}_{t-k}). \tag{2}$$

where N is the number of stocks available at time t.[1] This forms a long-short portfolio strategy with zero net investment. Importantly, this strategy involves no hindsight bias

[1] We assign zero weight to stocks at time t that lack historical returns at time t-k and suppress the notational dependence of N on t and k.

because it is based only on information available at the end of month t-k, strictly before month t. The realized return on this strategy at time t is

$$\pi_t(k) \;=\; \frac{1}{N}\sum_{i=1}^{N} w_{it}(k) r_{it} \;=\; \frac{1}{N}\sum_{i=1}^{N}(r_{it-k}-\bar{r}_{t-k}) r_{it}. \tag{3}$$

Substituting the single-factor model of equation (1) into equation (3) decomposes the average returns into three components

$$E[\pi_t(k)] \;=\; \sigma^2_{\mu} + \sigma_b^{2}\mathrm{Cov}(f_t,f_{t-k}) + \overline{\mathrm{Cov}}(e_{it},e_{it-k}), \tag{4}$$

where σ^2_{μ} denotes the cross-sectional variance of expected returns, σ_b^2 denotes the cross-sectional variance of factor sensitivities, and $\overline{\mathrm{Cov}}(e_{it},e_{it-k})$, denotes the average lag-k autocovariance of idiosyncratic returns. In other words returns to weighted relative strength strategies may stem from cross-sectional variation in returns, from systematic autocorrelation in market factors, or from autocorrelation in idiosyncratic stock returns.

We do not know whether autocorrelation in stocks returns is zero at short horizons. But the autocorrelation should vanish at long lags

$$\lim\,(k\rightarrow\infty)\;\; \mathrm{Cov}(f_t,f_{t-k}) \;=\; 0, \tag{5}$$

$$\lim\,(k\rightarrow\infty)\;\; \overline{\mathrm{Cov}}(e_{it},e_{it-k}) \;=\; 0.$$

In particular the "average" autocorrelation effect should be zero at long lags.[2] A simple consistent estimator of the cross-sectional variance of expected returns is obtained by averaging the weighted relative strength returns $\pi_t(k)$ at all lags. Our estimator of σ^2_{μ} is

$$\hat{\sigma}^2_{\mu} \;=\; \frac{1}{T(T-1)/2}\sum_{t=2}^{T}\sum_{k=1}^{T-1}\pi_t(k) \;=\; \frac{1}{NT(T-1)/2}\sum_{t=2}^{T}\sum_{k=1}^{T-1}\sum_{i=1}^{N}(r_{it-k}-\bar{r}_{t-k})(r_{it}-\bar{r}_t). \tag{6}$$

[2] Typical mixing conditions for stationary processes require the dependence to decay as an exponential function of the lag k.

The expectation of the estimator $\hat{\sigma}^2_\mu$ equals the true cross sectional variance plus terms due to autocorrelation.

$$E[\hat{\sigma}^2_\mu] = \sigma^2_\mu + \frac{1}{T(T-1)/2} \sum_{t=2}^{T} \sum_{k=1}^{T-1} (Cov(f_t, f_{t-k}) + \overline{Cov}(e_{it}, e_{it-k})) \qquad (7)$$

This autocorrelation effect should be quite small since we average over all lags up to 40 years. In fact Conrad and Kaul (1998) and Jegadeesh and Titman (2002) use very similar bootstrap estimators based on randomly sampling the dates t and lags k for individual stock returns in equation (6).[3] The essential difference is that our estimator takes an equally weighted average over *all* stock pairs of dates and lags, whereas Conrad and Kaul (1998) and Jegadeesh and Titman (2002) use an equally weighted average over 500 *randomly selected* combinations of date and lag. To facilitate comparison of our results with those previous bootstrap results we calculate standard errors by dividing the population standard deviation of $\pi_t(k)$ by the square-root of 500.

This paper uses monthly returns over the years 1963-2002. Our sample includes NYSE/AMEX-listed firms whose data is available on the CRSP monthly returns file.[4] This time period is comparable to the time period used for related research and allows us to consider COMPUSTAT data without substantially changing the sample period.

Table 1 shows our estimate of the cross-sectional variance of returns based on equation (6). The null hypothesis is that σ^2_μ is zero, i.e., there is no cross-sectional variation in expected stock returns. This hypothesis is rejected if the estimate is significantly positive. The estimate is actually slightly negative, small, and statistically

[3] Jegadeesh and Titman (2002) made the crucial distinction of bootstrap sampling returns without replacement to avoid forming a portfolio using the same return as the holding period month. Our estimator also avoids this because it corresponds to a lag of k=0.
[4] In comparison, Conrad and Kaul (1998) use NYSE/AMEX-listed firms for the period 1926-1989, while Jegadeesh and Titman (2002) use NYSE/AMEX-listed firms for the period 1965-1997.

insignificant at -0.0009% per month. This agrees with the statistically insignificant results found by Jegadeesh and Titman (2002) using 6-month returns. Theoretically buying historical winners and selling historical losers should invest in stocks with high average returns. But Table 1 shows it is not profitable to buy winners and sell losers based on a random, typically distant, historical date.

1.2 A Seasonal Model of Stock Returns

Inspired by the literature on seasonality in asset pricing models, we now generalize the model in equation (1). We allow the individual means to be time-dependent

$$r_{it} - \overline{r}_t = \mu_{it} + b_i f_t + e_{it}, \tag{8}$$

where the mean depends on the calendar month

$$\mu_{it} = \mu_{it+12}. \tag{9}$$

This means particular stocks may have distinctly high or low expected returns every 12 months. For example, Keim (1983) found small stocks had high returns in January. This model has distinct implications for the performance of weighted relative strength strategies at annual lags. It predicts the component of cross-sectional variance σ^2_μ in monthly weighted relative strength returns $\pi_t(k)$ will be positive when k is a multiple of 12. In other words it predicts a positive return to buying winners and selling losers based on their historical performance in the same calendar month of any previous year.

Table 1 reports positive results when averaging $\pi_t(k)$ only over annual lags k = 12, 24, … 468. This is quite surprising because Table 1 reports an average over all annual lags in our sample of 40 years. The point estimate of σ^2_μ is 1.33 basis points which seems small. But it implies large variation in stock returns. The equivalent annualized standard deviation of expected stock returns is $12\sqrt{0.000133} = 13.8\%$. It is instructive to compare this with conventional assessments of the cross-sectional variance of expected stock returns. Jegadeesh and Titman (2002) consider Fama and French (1992) estimates of the Capital Asset Pricing Model as a benchmark. The cross-sectional standard deviation of beta is 0.31. With a 6% market risk premium this corresponds to a 1.86% standard deviation of annual stock returns. Our estimate is over seven times as large.

Our seasonal model does not restrict the cross-sectional variance to be identical across different months. The evidence of Keim (1983) and Tinic and West (1984) using small stocks and high-beta stocks suggests the variation in expected return may be higher in January. Therefore Table 2 presents results separately for each calendar month. Each row of Table 2 is equivalent to deleting all but one calendar month from every year in the dataset, and then calculating the estimate of cross-sectional variance using only that one remaining month from every year. The estimated cross-sectional variance in January is quite high at .000877. This corresponds to a remarkably high annualized standard deviation of $12\sqrt{.000877} = 35.5\%$ per year. However, seasonality is not limited to January, the effect is positive in all twelve calendar months. In addition the estimates for October and December are also slightly above average. In principle it should be possible to exploit the large cross-sectional differences by buying stocks with large seasonal returns and selling stocks with negative seasonal returns.

2. Seasonal Portfolio Strategies

The estimates of the previous section indicate large seasonal cross-sectional variation in expected stock returns. These estimates are based on positive average returns to weighted relative strength strategies based on historical returns of lag k when k is a multiple of 12. The magnitudes of profits from weighted relative strength strategies are difficult to interpret because the absolute dollar portfolio weights vary through time. A simpler and highly correlated strategy involves buying an equally weighted long position of stocks in the top decile of returns for month t-k, and selling $1 short of stocks in the bottom decile of returns for month t-k. We construct these winner-loser portfolios from a sample that includes all stocks with available return data in the formation month t-k preceding the event month t. This avoids any sample selection or "look-ahead" bias in our strategies. We then compute average returns on these strategies for different lags. Figure 1 shows the average returns of winner-loser decile spreads as a function of lag k. It also shows a smoothed version of this graph. The smoothed graph is perfectly consistent with previous literature. It shows a positive "momentum" effect for the first year (Jegadeesh (1990), Jegadeesh and Titman (1993, 2001)), and then a negative effect from 2-7 years (DeBondt and Thaler (1985, 1987)). But the unsmoothed graph reveals a distinct periodic pattern. There is a positive "pulse" at annual intervals from 12 to 240 months. This corresponds to the effect of owning stocks with high expected returns in seasonal months. Jegadeesh (1990) noticed this effect at horizons of 24 months and 36 months. The important distinction is Jegadeesh presented it in the context of temporary returns to short-term (1-12 month) stock momentum, while we recognize it as a separate, permanent seasonal effect of expected returns.

The periodic pattern in Figure 1 motivates us to try portfolio strategies based on seasonal historical return. We wish to form winner-loser strategies to exploit the effect of lagged returns at distinct annual intervals. In order to distinguish results over the Jegadeesh and Titman horizon of 1 year from the DeBondt and Thaler horizon of 2-5 years, we consider these intervals separately. We further consider portfolio formation intervals for years 6-10, years 11-15, and years 16-20. We form decile portfolios based on the average monthly return of stocks over all months in each lagged interval and measure the returns over the next month. For example, the 2-5 year winner portfolio held in January 2003 would be an equally weighted combination of stocks that had the highest average return from January 1998 to December 2001. To exploit seasonal variation we also divide the historical formation months into annual lags 12, 24, … , 240 and non-annual lags. Then we form decile spreads based on average performance in only those lags. The essential difference is instead of sorting stocks based on their returns in all contiguous months in a given historical interval, we sort on noncontiguous months.

Table 3 shows the results for portfolios based on "all" months in a given interval reproduce the findings of previous literature. The year 1 returns are uniformly increasing through the deciles, with the decile 10 winners outperforming the decile 1 losers by 146 basis points per month. Yet the year 2-5 winners underperform the losers by 107 basis points per month. Interestingly the winners also slightly underperform the losers with formation lags of 6-10, 11-15, and 16-20 years.

Table 3 also shows the returns to winner-loser strategies formed only on the basis of only the annual or non-annual months in the lagged time interval. In the year 1 interval there is only one annual lag – the 12-month lagged return. Given the decile spreads based on "all" months in the 1-year interval are positive, it is not surprise to see the decile spread is still positive when sorting only on the 12-month lagged return. But the magnitude of 115 basis points is quite impressive compared to the 146 basis points for all months. It appears most of the performance from winners and losers over the past year can be captured based on stocks that were winners or losers exactly 12 months ago. In fact the t-statistic for this 12-month winner-loser strategy exceeds the t-statistic for the full-year strategy, indicating it produces a higher Sharpe ratio. The simple strategy of buying winners and selling losers based on the 12-month lagged returns produces a higher return for risk than the full-year strategy.

The surprising results come from the longer horizon strategies. Recall winner-loser deciles based on "all" months in years 2-5 lost 107 basis points per month. But when sorting only on annual lags 24, 36, 48, and 60 the decile 1 winners outperformed decile 10 losers by 67 basis points. We get a positive effect in the middle of DeBondt and Thaler's negative effect. This positive effect appears for all the annual strategies – 68 basis points based on years 6-10, 66 basis points based on years 11-15, and 52 basis points based on years 16-20. All these returns are statistically significant at conventional levels.

Table 3 shows the average winner-loser returns based on non-annual months are negative for all formation intervals and usually statistically significant. It is striking that winner-loser strategies can produce significant returns up to 20 years later, and that these returns are positive in some months and negative in adjacent months.

If our explanation in the previous section is correct then the positive returns to annual winner-loser strategies are a result of buying stocks with high average returns and selling stocks with low average returns. This effect should be eliminated subtracting the average returns from each stock. Following Jegadeesh and Titman (2002) we consider three different types of estimates of mean return for each stock. The first type of estimate is the average monthly return on a stock prior to the formation month t-k. We call this the "preranking estimate".

$$\hat{\mu}_i(1) = \frac{1}{t-k-1}\sum_{j=1}^{t-k-1} r_{ij}.$$

The second type of estimate uses all months after the formation month t-k except for the holding month t.

$$\hat{\mu}_i(2) = \frac{1}{T-t+k-1}\sum_{j>t-k, j\neq t}^{T} r_{ij}.$$

We call this the "postranking estimate". Finally, we also compute average returns over all months excluding the formation month t-k and the holding month t.

$$\hat{\mu}_i(3) = \frac{1}{T-2}\sum_{j\neq t-k, j\neq t} r_{ij}.$$

Alternatively we can estimate the average returns for each stock using only those preranking or postranking months that match the calendar month of the holding period t. For example, if the holding period is March 2000 then we would compute the seasonal preranking estimate using average return for each stock based on all March returns prior to the ranking month. Similarly we form seasonal postranking and seasonal combined estimates of average return for each stock in each month.

Table 4 presents the seasonal profits of Table 3 adjusted by the different estimates of expected return. It shows the adjustment to mean returns using "all" months does not substantially alter the profitability of annual strategies. And it does not appear to matter whether the adjustment uses preranking or postranking months. But adjusting the means using seasonal estimates sharply reduces the magnitude of the one year annual effect and eliminates the statistical significance of profits beyond one year. It appears the profitability of these strategies is entirely explained by seasonal variation in individual stock returns.

The results in Table 2 of the previous section indicate the seasonal variation in mean returns is not equal in all months. In this case the profits to seasonal strategies should be higher in months with large variance in mean returns – January, October, and December. Table 5 addresses this by separately reporting results of Table 3 for each calendar month. It shows the average decile spread returns for annual seasonal strategies in each month. For example, the Years 2-5 January return of 3.89% means that stocks in the upper decile of return in January of four previous years (lagged 2-5 years) outperformed the lowest decile stocks by almost 4% over the holding period January. The Years 2-5 annual strategy averaged over 1% in the months of November and December. The other annual strategies also earned particularly high returns in January and tended to have above average performance in October and November. The seasonal effect seems to vary across calendar months and be particularly strong in the last few month and the first month of the calendar year.

3. Seasonal Returns across Size and Industry

It is not clear why stocks would earn substantially different expected returns in one month than another. One explanation is that seasonal variation in returns is associated with stocks have systematic seasonal risks. These risks might be correlated with factors such as size or industry. This section considers the performance of annual winner-loser strategies within size and industry categories. We implement the strategies within three size-based subsamples (small, medium, and large) and decompose the profits into inter-industry and intra-industry components. Measuring the profitability of winner-loser strategies within size subsamples shows whether the performance of our strategies is limited to small or large firms. Size and industry membership also provide diagnostic information about the source of the winner-loser profits. Size may proxy for stock market liquidity. Size and industry may be considered as characteristics that are related to expected return (Daniel and Titman (1997)). In this case the cross-sectional dispersion of expected returns should be less within size subsamples or industries than within the full sample. Therefore if the winner-loser profits are related to expected returns then they will be reduced by controlling for size and industry.

Table 7 shows the winner-loser profits for different strategies within size subsamples. Interestingly the years 2-5 and years 6-10 annual strategies are more profitable among large firms than among small or medium firms. In contrast the years 2-5 and years 6-10 non-annual strategies are most profitable among small firms. The year 11-15 annual winner-loser profits remain statistically significantly positive in all size categories, and the year 16-20 annual winner loser profits remain significant within medium and large firms. The profitability and pattern of winner-loser profits is not limited to a particular size subsample.

Previous research by Moskowitz and Grinblatt (1999) and Lewellen (2002) has shown a role for industry in 1-year winner-loser strategies. We address this by classifying stocks into the 20 industry groups of Grinblatt and Moskowitz and then decomposing returns into industry and intra-industry components. The industry component of a stock return is simply the return of its equally weighted industry group, and the intra-industry of a stock is its return in excess of industry. Thus our winner-loser strategies have industry and intra-industry components that sum to the whole return by definition.

Figure 2 shows the average monthly winner-loser decile spreads of Figure 1 broken into intra-industry and inter-industry components. The inter-industry graph in Panel A has the same magnitude and annual pattern in event time as the whole return in Figure 1. Panel B shows the industry component is positive for event times less than 12 months, consistent with the findings of Moskowitz and Grinblatt (1999) and Grinblatt and Moskowitz (2003). But the industry component is small, with no periodic pattern in event time. The "industry momentum" effect appears to be distinct and qualitatively different phenomenon.

Table 7 decomposes decile spreads of annual and non-annual winner-loser strategies into inter-industry and intra-industry components. The magnitudes and statistical of significance of intra-industry components are similar to the raw returns in Table 1. While the industry component is occasionally statistically significant it is always comparatively small in magnitude. For example, the industry component of the years 6-10 annual strategy is 12 basis points per month, but the inter-industry component is 58 basis points per month. Neither industry nor size explains the bulk of the winner-loser returns. It appears the seasonal variation in expected returns is not subsumed by size nor industry categories.

4. Stock Returns in Earnings Announcement Months

If seasonal expected stock returns are not related to size nor industry then they may not be related to systematic risk. Instead they may be related to firm-specific characteristics. If markets have biased expectations about firm-specific information then winner-loser strategies may be exploiting predictable returns associated with dissemination of earnings. This section examines the returns of past winner and losers over their quarterly earnings announcement months.

Consider, for example, the possibility that markets underreact to persistent seasonal earnings information. In this case stocks with favorable earnings news in one year will tend to have favorable news in subsequent years. And incomplete market reaction in the first year will result in positive returns around under-anticipated earnings announcements in subsequent years (see Bernard and Thomas (1990)). To the extent that firms consistently announce earnings in the same calendar month, this may explain the success of annual winner-loser strategies.

To investigate the relation of our results to earnings announcements, we limit the data to firms covered by the COMPUSTAT earnings database. This results in 6,631 firms over the period January 1965 to December 2002. Roughly one-third of the observations have an earnings announcement and two-thirds do not have an earnings announcement in any given month.

Table 8 separates the monthly holding periods for winner-loser strategies into earnings announcement months and non-announcement months. The formation of winner and loser deciles is identical to the whole sample in Table 3, but is limited those firms covered by Compustat. In any given month roughly one third of the firms have earnings announcements. We construct decile portfolios using all firms, using only those firms with earnings announcements in that month, and using only firms without announcements. The winner-loser returns over the whole sample are quite similar to the returns over the larger sample. For example the average winner-loser decile spread for the year 1 non-annual strategy was 110 basis points for the Compustat sample, and 117 basis points for the larger sample in Table 3. The year 1 non-annual results confirm Jegadeesh and Titman's (1993) finding that positive returns are concentrated around earnings announcements. The average winner-loser decile spread of 191 basis points is more than twice the 91 basis point return in non-announcement months. This is consistent with markets underreacting to previous persistent earnings news and then completing the reaction when new earnings are released in the holding month.

But the results for longer horizons are not very different across announcement months and non-announcement months. The years 2-5 annual winners outperform years 2-5 annual losers by 78 basis points in earnings announcement months, and 67 basis points in non-announcement months. And years 2-5 non-annual winners continue to underperform corresponding losers by over 100 basis points in both announcement and non-announcement months. The results for longer horizons are qualitatively similar to results for years 2-5 strategies. Annual winners outperform losers in announcement months and non-announcement months, while non-annual winners underperform losers.

Overall our results are consistent with those of previous research, yet also demonstrate some distinctive new features of stock returns. We confirm Jegadeesh and Titman's (1993) finding that positive returns to year 1 winner-loser strategies are concentrated around earnings announcements. But the longer horizon strategies show little difference in announcement months. The positive returns to annual winner-loser strategies also contrast with findings of Bernard and Thomas (1990). Bernard and Thomas found returns around earnings announcements to be negatively related to earnings surprise 4 quarters earlier. This would suggest winners would underperform losers at subsequent annual intervals, which is the opposite of our results. Positive returns to annual winner-loser strategies are not associated with earnings and represent a distinct anomaly from previous literature.

5. Discussion

This section discusses the interpretation of our results in terms of economic theories that could explain seasonal risk premia across individual stocks.

5.1 Risk-Adjusted Returns

To the extent that market efficiency does not eliminate seasonal differences in expected returns, there may be some offsetting seasonal features of these stocks. We attempted to use size and industry as proxies for risk and found little change in results. Unreported diagnostics show little sensitivity of seasonal strategies to beta or to the Fama and French (1993) three factors. This is similar to results for the momentum anomaly (see Fama and French (1996) and Grundy and Martin (2001)). Therefore systematic risk does not seem to be an empirically successful explanation for seasonal variation in expected returns.

5.2 Returns and Volume

The monthly returns of stocks may be affected by seasonal changes in the stocks' volume. A related paper by Lee and Swaminathan (2000) has investigated the interaction of intermediate horizon momentum returns and volume. In unreported exploration we have found seasonal patterns in individual stock trading volume at both annual and quarterly frequencies. While this paper has focused on the annual seasonal pattern in returns, closer inspection of Figure 1 reveals a quarterly pattern of positive returns to winner-loser strategies as a function of monthly historical lag within the first year. The quarterly and annual patterns in expected return may be related to similar patterns individual stock volume. This suggests a seasonal pattern of liquidity may help explain expected returns.

5.3 Transaction Costs

The estimates of our paper have simple interpretations in terms of portfolio returns. This suggests they might be captured through an investment strategy. For example, the magnitude of the annual decile spread strategies in Table 3 exceeds 60 basis points per month, which is comparable to many other “anomalies” in the empirical finance literature. However, there is an important distinction between seasonal strategies and simple “momentum” or “contrarian” strategies of Jegadeesh and Titman (1993) or DeBondt and Thaler (1985, 1987). Momentum and contrarian strategies require rebalancing only once or twice per year. In contrast seasonal strategies require rebalancing the entire portfolio every month. In this respect it may not be generally profitable to incur round trip transactions costs for a 60-70 basis point monthly gain. Compared to typical transactions costs, returns to seasonal strategies are only particularly large in January. However, Sadka (2001) and Korajczyk and Sadka (2004) show evidence that transaction costs may also be higher in January. In any case the existence of short-lived fluctuations in monthly expected return may not form an effective foundation for a long-term investment strategy. On the other hand the results are strong enough to merit investment consideration for an active portfolio. It is relatively simple to postpone the sale or purchase of a particular stock if it has a large positive or negative expected return over the next month.

5.4 Behavioral Explanations

It is tempting to conjecture other variables that may cause annual patterns in expected returns. There is a substantial body of behavioral research that explains return variation in terms of underreaction and overreaction to news (DeBondt and Thaler (1985), Barberis, Shleifer, and Vishny (1998), Daniel, Hirshleifer, Subrahmanyam (1998), and Hong and Stein (1999)). While these theories are typically framed in terms of temporal reaction, they may apply to our current findings if underreaction and overreaction occur in response to regular seasonal news. In order to fit the data, this reaction must be both seasonal and abruptly discontinuous. For example, Table 1 and Figure 1 show the average decile spread to buying stocks based on their 12-month historical return is 115 points per month. In contrast the decile spread to buying stocks based on their 13-month historical return is less than -30 basis points per month, and the average return to buying stocks based on their 14-month historical return is -50 basis points per month. Any theory that successfully explains the annual pattern of expected returns must explain both the seasonality and the sharp difference between returns in one month and returns in adjacent months.

6. Conclusions

This paper investigates seasonality in the cross-section of expected stock returns. It uses the Conrad and Kaul (1998) and Jegadeesh and Titman (2002) methodology of buying stocks with high past returns and selling stocks with low past returns. Intuitively if performance is due to high expected returns then these strategies will invest in stocks with above average returns and should have positive returns in the future. In fact these strategies are not profitable when investing in all months. But they are profitable when forming seasonal portfolios based on stock returns and then holding those portfolios in future years over the same calendar months.

Our empirical results are important for asset pricing because they provide new and different evidence on the expected returns among common stocks. We estimate an annualized cross-sectional standard deviation of 13.8%, which is over seven times as large as conventional estimates of asset pricing models. Using the same estimation methodology without seasonal structure overlooks these large differences in expected return. In other words, consideration of seasonality completely changes the implications of stock returns for asset pricing models.

The results of our paper are quite robust. We formed a variety of seasonal strategies and found the positive performance continues for up to 20 years. It appears to be a permanent effect due to cross-sectional variation in expected returns, not a transitory effect caused by autocorrelation in returns. This performance is eliminated when demeaning stock returns by their seasonal averages. This confirms that returns are caused by cross-sectional dispersion in means. The estimated cross-sectional variance is positive in all twelve calendar months. It appears particularly high in January, but is statistically significant in other months and above average in October and December.

Additional diagnostics show the effect occurs within industry and size categories. In this respect it does not appear to be a byproduct of stock characteristics or simple proxies for risk. It is also not associated with earnings releases, a natural candidate for annual seasonal news. It remains to explain why individual stocks would exhibit strong monthly patterns in their expected returns. While we do not have the answer, we have noted similar cross-sectional patterns in trading volume. Future research can explore the patterns in volume and potentially provide a unified theory to jointly explain seasonal patterns in expected return in terms of underlying variables that influence trading.

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Table 1

Weighted Relative Strength Strategy (WRSS) Profits and Cross-Sectional Dispersion in Mean Returns

For every pair of month t and its lag month k we calculate the return of each stock in each month excess of the equally weighed market return that month (using the cross-section of stocks with available returns in both months j and k). For each stock, the product of is excess returns during months t and k is calculated. The sum of these products across firms is denoted $\pi_k(t)$ and it represents a return of a zero cost portfolio strategy. Three groups of returns are then defined: "All" includes all $\pi_k(t)$ such that $k\neq0$; "Non-Annual" include all $\pi_k(t)$ such that $k\neq12i$, for any integer i; and "Annual" includes all $\pi_k(t)$ such that $k=12i$, for any non-zero integer i. The table reports the mean return of each group. Standard errors are calculated by dividing the population standard deviation of each group by the square root of 500. The analysis employs NYSE/AMEX-listed firms for the period January 1963 through December 2002.

	All	Non-Annual	Annual
Mean (%)	-0.0006	-0.0019	0.0133
T-statistic	-0.26	-0.65	3.57

Table 2

Seasonal Cross-Sectional Dispersion in Mean Returns

For every pair of month t and its lag month k we calculate the return of each stock in each month excess of the equally weighed market return that month (using the cross-section of stocks with available returns in both months j and k). For each stock, the product of is excess returns during months t and k is calculated. The sum of these products across firms is denoted $\pi_k(t)$ and it represents a return of a zero cost portfolio strategy. A group of returns is formed for each calendar month by including all $\pi_k(t)$ such that both t and k equal that particular calendar month (and also $k\neq0$). The table reports the mean return of each group. Standard errors are calculated by dividing the population standard deviation of each group by the square root of 500. The analysis employs NYSE/AMEX-listed firms for the period January 1963 through December 2002.

January	Mean (%)	0.0877
	T-statistic	11.47
February	Mean (%)	0.0036
	T-statistic	1.08
March	Mean (%)	0.0049
	T-statistic	1.78
April	Mean (%)	0.0085
	T-statistic	3.35
May	Mean (%)	0.0026
	T-statistic	1.14
June	Mean (%)	0.0052
	T-statistic	2.26
July	Mean (%)	0.0013
	T-statistic	0.49
August	Mean (%)	0.0015
	T-statistic	0.56
September	Mean (%)	0.0072
	T-statistic	2.78
October	Mean (%)	0.0163
	T-statistic	4.13
November	Mean (%)	0.0061
	T-statistic	1.57
December	Mean (%)	0.0149
	T-statistic	5.19

Table 3
Returns of Strategies Based on Past Performance

Every month stocks are grouped into ten portfolios (with equal number of stocks in each portfolio) according to various categories based on past performance. For example, the trading strategy that is formed based on past annual returns during years 2 through 5 ranks stocks according to their average returns during the historical lags 24, 36, 48, and 60. The stocks in each portfolio are assigned equal weight, and the portfolios are rebalanced monthly. The average monthly returns of the various trading strategies for the period January 1965 through December 2002 (456 months) are reported below, as well as the corresponding *t*-statistics (two digit numbers). The analysis uses NYSE/AMEX-listed stocks.

Strategy		1 (losers)	2	3	4	5	6	7	8	9	10 (winners)	10-1
Year 1	All	0.0064	0.0090	0.0101	0.0108	0.0117	0.0125	0.0136	0.0152	0.0173	0.0210	0.0146
		1.65	3.12	3.97	4.54	5.11	5.50	5.77	6.05	6.31	6.39	5.58
	Annual	0.0082	0.0097	0.0106	0.0111	0.0121	0.0127	0.0138	0.0148	0.0160	0.0197	0.0115
		2.53	3.66	4.36	4.77	5.20	5.56	5.78	5.82	5.77	5.80	7.60
	Non-Annual	0.0080	0.0105	0.0111	0.0109	0.0119	0.0120	0.0128	0.0144	0.0165	0.0197	0.0117
		1.99	3.59	4.23	4.53	5.23	5.26	5.48	5.88	6.08	6.13	4.20
Years 2-5	All	0.0191	0.0145	0.0135	0.0127	0.0129	0.0126	0.0128	0.0116	0.0112	0.0084	-0.0107
		5.28	5.25	5.50	5.59	5.68	5.39	5.25	4.50	3.99	2.57	-5.02
	Annual	0.0110	0.0103	0.0105	0.0109	0.0126	0.0124	0.0137	0.0149	0.0157	0.0177	0.0067
		3.48	3.85	4.37	4.74	5.44	5.27	5.74	5.83	5.71	5.44	5.35
	Non-Annual	0.0199	0.0159	0.0141	0.0130	0.0129	0.0126	0.0120	0.0114	0.0101	0.0074	-0.0125
		5.42	5.67	5.78	5.74	5.71	5.36	4.99	4.44	3.59	2.28	-5.60
Years 6-10	All	0.0156	0.0147	0.0138	0.0131	0.0121	0.0121	0.0111	0.0111	0.0118	0.0117	-0.0039
		5.17	5.76	5.97	5.82	5.40	5.27	4.71	4.46	4.31	3.68	-3.32
	Annual	0.0104	0.0093	0.0101	0.0112	0.0117	0.0114	0.0135	0.0144	0.0155	0.0172	0.0068
		3.44	3.68	4.27	4.90	5.24	5.10	5.90	6.02	5.92	5.60	6.15
	Non-Annual	0.0167	0.0156	0.0146	0.0135	0.0126	0.0114	0.0110	0.0102	0.0105	0.0112	-0.0055
		5.47	6.14	6.23	5.96	5.61	5.02	4.64	4.09	3.91	3.51	-4.62
Years 11-15	All	0.0135	0.0116	0.0110	0.0120	0.0120	0.0122	0.0123	0.0126	0.0127	0.0133	-0.0002
		4.67	4.76	4.87	5.46	5.60	5.51	5.54	5.33	5.01	4.69	-0.17
	Annual	0.0101	0.0104	0.0102	0.0111	0.0112	0.0129	0.0130	0.0134	0.0145	0.0166	0.0066
		3.61	4.25	4.52	5.13	5.24	5.87	5.83	5.77	5.81	5.97	6.43
	Non-Annual	0.0144	0.0122	0.0122	0.0120	0.0122	0.0115	0.0124	0.0121	0.0119	0.0125	-0.0019
		4.98	5.02	5.28	5.37	5.61	5.37	5.56	5.07	4.71	4.45	-1.77
Years 16-20	All	0.0150	0.0117	0.0116	0.0110	0.0113	0.0120	0.0117	0.0113	0.0123	0.0119	-0.0031
		5.48	4.89	5.21	5.11	5.29	5.54	5.43	5.01	5.14	4.65	-2.82
	Annual	0.0100	0.0100	0.0103	0.0111	0.0112	0.0115	0.0122	0.0131	0.0137	0.0153	0.0052
		3.78	4.24	4.61	5.02	5.14	5.36	5.64	5.95	5.86	5.84	4.58
	Non-Annual	0.0154	0.0128	0.0115	0.0122	0.0115	0.0111	0.0108	0.0120	0.0109	0.0115	-0.0039
		5.57	5.38	5.16	5.60	5.37	5.17	4.95	5.37	4.61	4.49	-3.35

Table 4
Mean-Adjusted Returns of Strategies Based on Past Performance

Every month stocks are grouped into ten portfolios (with equal number of stocks in each portfolio) according to past performance during various ranking periods. For example, the trading strategy that is formed based on past annual returns during years 2 through 5 ranks stocks according to their average returns during the historical lags 24, 36, 48, and 60. The stocks in each portfolio are assigned equal weight, and the portfolios are rebalanced monthly. For each stock each month, the monthly return is adjusted by subtracting off the stock's mean return. Three different periods are separately used to calculate this mean return: pre-ranking, post-ranking, and all. The pre-ranking period corresponds to the period starting at the beginning of the sample period until one month prior to the beginning of the ranking period. The post-ranking period includes all months one month after the end of the ranking period until the end of the sample period (excluding the return month itself). "All" includes all month during the pre- and post-ranking periods. The standard mean return corresponds to the regular monthly return average over the period, while to only returns during the same calendar month as the return month are used to calculate the seasonal mean return over the period. The table reports the average monthly adjusted returns of the long-short strategy (10 minus 1), as well as the corresponding *t*-statistics (two digit numbers). The analysis uses NYSE/AMEX-listed stocks for the period January 1965 through December 2002.

	No Adjustments	Standard Mean Adjustments			Seasonal Mean Adjustments		
Strategy		Pre-Ranking	Post-Ranking	All	Pre-Ranking	Post-Ranking	All
1 year	0.0115	0.0112	0.0119	0.0121	0.0063	0.0040	0.0070
	7.60	7.06	7.56	7.88	4.14	2.75	4.89
2-5 years	0.0067	0.0111	0.0082	0.0083	0.0038	-0.0014	0.0004
	5.36	6.62	6.54	6.58	2.28	-1.09	0.35
6-10 years	0.0069	0.0108	0.0081	0.0082	0.0018	0.0004	0.0010
	6.22	6.78	7.36	7.42	1.10	0.39	0.92
11-15 years	0.0065	0.0107	0.0075	0.0076	0.0008	0.0010	0.0010
	6.37	6.49	7.36	7.39	0.47	0.94	0.95
16-20 years	0.0051	0.0091	0.0058	0.0058	-0.0013	-0.0002	-0.0005
	4.43	5.59	5.06	5.09	-0.74	-0.21	-0.43

Table 5
Seasonal Returns of Strategies Based on Past Performance

Every month stocks are grouped into ten portfolios (with equal number of stocks in each portfolio) according to various categories based on past performance. For example, the trading strategy that is formed based on past annual returns during years 2 through 5 ranks stocks according to their average returns during the historical lags 24, 36, 48, and 60. The stocks in each portfolio are assigned equal weight, and the portfolios are rebalanced monthly. The average monthly returns of the various trading strategies are reported separately for every calendar month during the period January 1965 through December 2002. The corresponding *t*-statistics are also reported below (two digit numbers). In a separate column, returns are computed using all non-January months. The analysis uses NYSE/AMEX-listed stocks.

Strategy		January	February	March	April	May	June	July	August	September	October	November	December	Feb-Dec
Year 1	All	-0.0449	0.0101	0.0134	0.0217	0.0082	0.0307	0.0127	0.0126	0.0241	0.0189	0.0256	0.0419	0.0200
		-2.92	1.06	2.15	3.57	1.19	4.97	1.92	2.10	3.34	1.99	2.60	5.07	8.62
	Annual	0.0333	0.0019	0.0093	0.0099	0.0049	0.0110	0.0006	0.0083	0.0130	0.0097	0.0153	0.0209	0.0095
		3.52	0.32	2.17	3.17	1.41	3.16	0.14	2.22	3.27	1.80	2.70	3.84	6.93
	Non-Annual	-0.0683	0.0090	0.0091	0.0244	0.0075	0.0295	0.0138	0.0133	0.0224	0.0189	0.0214	0.0397	0.0190
		-3.98	0.96	1.53	4.08	1.09	4.84	1.98	2.10	2.99	1.95	2.23	4.86	8.18
	Difference	0.1016	-0.0072	0.0002	-0.0145	-0.0026	-0.0186	-0.0132	-0.0050	-0.0093	-0.0092	-0.0061	-0.0188	-0.0095
		4.77	-0.62	0.03	-2.04	-0.39	-3.45	-1.78	-0.73	-1.23	-0.88	-0.59	-2.40	-3.84
Years 2-5	All	-0.0807	-0.0237	-0.0189	-0.0059	-0.0085	-0.0083	-0.0107	0.0045	-0.0004	0.0091	0.0045	0.0107	-0.0043
		-6.19	-3.22	-3.60	-1.11	-1.42	-1.87	-2.05	0.90	-0.11	1.72	1.04	1.65	-2.56
	Annual	0.0389	-0.0032	0.0030	0.0043	0.0020	0.0013	0.0025	0.0006	0.0040	0.0028	0.0104	0.0137	0.0038
		6.16	-0.83	0.87	1.11	0.67	0.44	0.76	0.16	1.13	0.60	2.23	3.39	3.31
	Non-Annual	-0.0960	-0.0234	-0.0182	-0.0071	-0.0084	-0.0093	-0.0108	0.0057	-0.0034	0.0098	0.0046	0.0069	-0.0049
		-7.30	-3.19	-3.40	-1.44	-1.41	-2.08	-2.06	1.07	-0.90	1.65	1.11	1.11	-2.89
	Difference	0.1350	0.0203	0.0212	0.0113	0.0104	0.0106	0.0133	-0.0051	0.0075	-0.0070	0.0058	0.0068	0.0087
		9.33	2.82	3.68	2.38	1.68	2.21	2.31	-0.85	1.68	-0.81	1.07	1.24	4.74
Years 6-10	All	-0.0143	-0.0045	-0.0016	-0.0042	-0.0055	-0.0022	-0.0092	-0.0039	-0.0067	0.0017	0.0073	-0.0034	-0.0029
		-2.52	-0.99	-0.44	-1.09	-1.56	-0.73	-1.83	-1.21	-2.13	0.46	1.70	-0.96	-2.54
	Annual	0.0323	0.0008	0.0043	0.0025	0.0008	0.0064	-0.0003	0.0005	0.0024	0.0106	0.0118	0.0091	0.0044
		4.98	0.23	1.17	1.10	0.26	2.20	-0.11	0.18	0.89	2.72	2.77	2.92	4.57
	Non-Annual	-0.0235	-0.0060	-0.0047	-0.0033	-0.0054	-0.0009	-0.0091	-0.0051	-0.0072	0.0002	0.0051	-0.0062	-0.0039
		-3.88	-1.28	-1.33	-0.83	-1.49	-0.32	-1.73	-1.62	-2.20	0.07	1.39	-1.73	-3.37
	Difference	0.0559	0.0068	0.0090	0.0057	0.0061	0.0074	0.0088	0.0056	0.0096	0.0104	0.0068	0.0154	0.0083
		6.54	1.15	1.90	1.38	1.39	1.87	1.55	1.31	2.14	2.05	1.71	3.69	5.96
Years 11-15	All	-0.0090	-0.0019	0.0029	-0.0011	-0.0005	0.0022	-0.0013	0.0008	-0.0012	0.0044	0.0092	-0.0065	0.0006
		-1.72	-0.46	0.81	-0.40	-0.15	0.76	-0.42	0.25	-0.44	0.99	3.23	-2.01	0.62
	Annual	0.0269	0.0025	0.0043	0.0005	0.0076	0.0025	-0.0003	0.0023	0.0089	0.0093	0.0054	0.0091	0.0047
		5.63	0.79	1.24	0.18	2.63	0.86	-0.11	0.77	3.23	2.18	1.34	2.61	4.82
	Non-Annual	-0.0220	-0.0054	0.0022	-0.0001	-0.0014	0.0020	-0.0015	0.0027	-0.0033	0.0016	0.0090	-0.0067	-0.0001
		-4.24	-1.35	0.66	-0.03	-0.44	0.70	-0.47	0.90	-1.26	0.31	2.89	-2.08	-0.08
	Difference	0.0489	0.0079	0.0021	0.0006	0.0089	0.0006	0.0013	-0.0004	0.0121	0.0077	-0.0035	0.0158	0.0048
		5.89	1.67	0.47	0.17	2.59	0.13	0.33	-0.09	2.87	1.08	-0.69	4.16	3.52
Years 16-20	All	-0.0148	-0.0116	-0.0038	-0.0058	0.0012	0.0006	-0.0060	-0.0042	0.0009	0.0001	0.0067	-0.0006	-0.0020
		-3.81	-3.14	-1.17	-1.75	0.34	0.18	-1.79	-1.13	0.26	0.03	1.58	-0.13	-1.79
	Annual	0.0297	-0.0036	0.0022	0.0044	0.0033	0.0064	0.0023	0.0023	0.0038	0.0084	0.0026	0.0009	0.0030
		4.23	-1.18	0.63	1.53	0.97	1.91	0.81	0.57	1.29	1.99	0.76	0.35	2.99
	Non-Annual	-0.0209	-0.0117	-0.0040	-0.0080	0.0028	-0.0014	-0.0046	-0.0044	0.0022	-0.0023	0.0059	0.0003	-0.0023
		-3.62	-3.05	-1.22	-2.20	0.83	-0.40	-1.40	-1.37	0.73	-0.52	1.50	0.08	-2.07
	Difference	0.0506	0.0081	0.0063	0.0124	0.0005	0.0078	0.0069	0.0068	0.0016	0.0108	-0.0034	0.0006	0.0053
		4.37	2.32	1.40	2.75	0.13	1.94	1.63	1.27	0.40	1.87	-0.75	0.15	3.93

Table 6
Controlling for Industry Effects

Every month stocks are grouped into ten portfolios (with equal number of stocks in each portfolio) according to various categories based on past performance. For example, the trading strategy that is formed based on past annual returns during years 2 through 5 ranks stocks according to their average returns during the historical lags 24, 36, 48, and 60. The stocks in each portfolio are assigned equal weight, and the portfolios are rebalanced monthly. The monthly return of every stock is then decomposed into intra- and inter-industry components. The intra-industry component is the monthly return of the stock excess of its industry group return, and the inter-industry component is the monthly return of the industry group itself. The industry groups (twenty groups) are formed according to the classification in Moskowitz and Grinblatt (1999). The average monthly returns of the various trading strategies (for both intra- and inter-industry components) for the period January 1965 through December 2002 (456 months) are reported below, as well as the corresponding *t*-statistics (two digit numbers). The analysis uses NYSE/AMEX-listed stocks.

Strategy		Intra-Industry	Inter-Industry
Year 1	All	0.0124	0.0022
		5.50	3.97
	Annual	0.0103	0.0012
		8.20	2.98
	Non-Annual	0.0097	0.0020
		3.96	3.71
Years 2-5	All	-0.0096	-0.0011
		-4.98	-2.54
	Annual	0.0063	0.0004
		5.79	1.16
	Non-Annual	-0.0113	-0.0012
		-5.58	-2.85
Years 6-10	All	-0.0037	-0.0003
		-3.75	-0.58
	Annual	0.0056	0.0012
		5.77	3.70
	Non-Annual	-0.0049	-0.0006
		-4.83	-1.51
Years 11-15	All	-0.0005	0.0003
		-0.57	0.86
	Annual	0.0056	0.0010
		6.19	2.99
	Non-Annual	-0.0020	0.0001
		-2.13	0.25
Years 16-20	All	-0.0030	0.0000
		-3.25	-0.08
	Annual	0.0046	0.0006
		4.68	1.76
	Non-Annual	-0.0035	-0.0003
		-3.57	-0.79

Table 7
Relative Strength Portfolio Returns for Different Size Groups

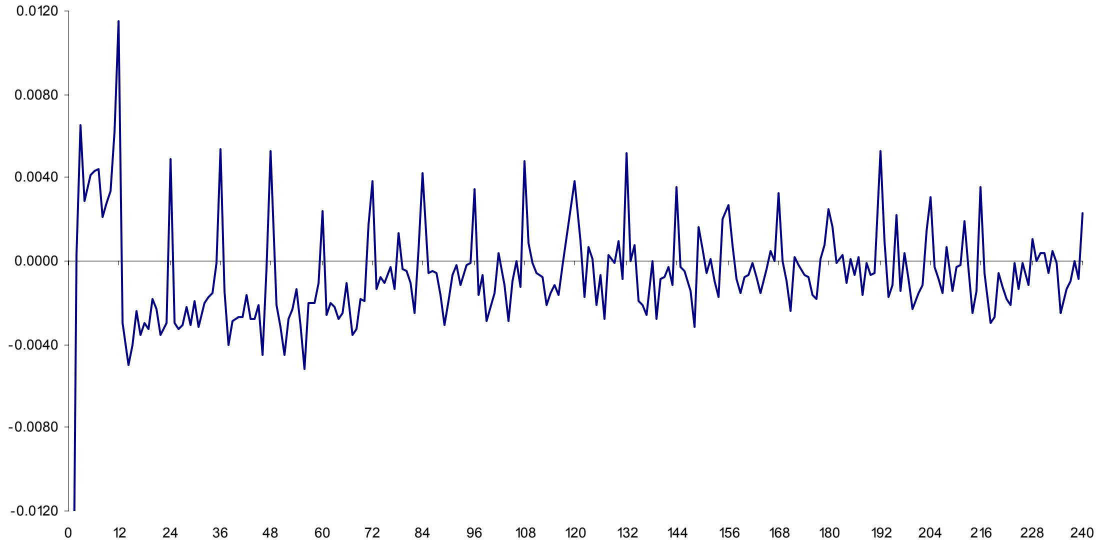
Every month stocks are grouped into ten portfolios (with equal number of stocks in each portfolio) according to various categories based on past performance. For example, the trading strategy that is formed based on past annual returns during years 2 through 5 ranks stocks according to their average returns during the historical lags 24, 36, 48, and 60. The stocks in each portfolio are assigned equal weight, and the portfolios are rebalanced monthly. The average monthly return difference between the highest past-performing decile and the lowest past-performing decile is then calculated for the period January 1965 through December 2002 (456 months). The table below reports the results of this procedure, performed separately for three different size groups (measured by market capitalization). Small firms are defined as the bottom 30 percent, large firms are the top 30 percent, and the remaining 40 percent are medium size firms. The size categorization is re-evaluated in the beginning of every month. The corresponding *t*-statistics are also reported (two digit numbers). The analysis uses NYSE/AMEX-listed stocks.

Strategy		Small	Medium	Large
Year 1	All	0.0129	0.0186	0.0102
		4.04	7.53	3.97
	Annual	0.0090	0.0095	0.0069
		4.27	6.68	4.19
	Non-Annual	0.0105	0.0173	0.0090
		3.07	6.86	3.49
Years 2-5	All	-0.0128	-0.0061	-0.0056
		-5.31	-3.78	-2.90
	Annual	0.0041	0.0042	0.0069
		2.32	3.54	5.01
	Non-Annual	-0.0161	-0.0071	-0.0079
		-6.34	-4.28	-4.11
Years 6-10	All	-0.0048	-0.0025	-0.0015
		-2.44	-1.85	-1.16
	Annual	0.0037	0.0045	0.0067
		1.82	3.57	5.90
	Non-Annual	-0.0059	-0.0035	-0.0032
		-2.84	-2.69	-2.47
Years 11-15	All	-0.0004	0.0005	0.0010
		-0.17	0.39	0.92
	Annual	0.0081	0.0034	0.0035
		3.17	2.67	3.34
	Non-Annual	-0.0012	-0.0006	-0.0007
		-0.47	-0.49	-0.61
Years 16-20	All	-0.0053	-0.0006	-0.0020
		-1.75	-0.39	-1.68
	Annual	0.0002	0.0035	0.0024
		0.06	2.34	2.17
	Non-Annual	-0.0037	-0.0013	-0.0024
		-1.14	-0.88	-2.05

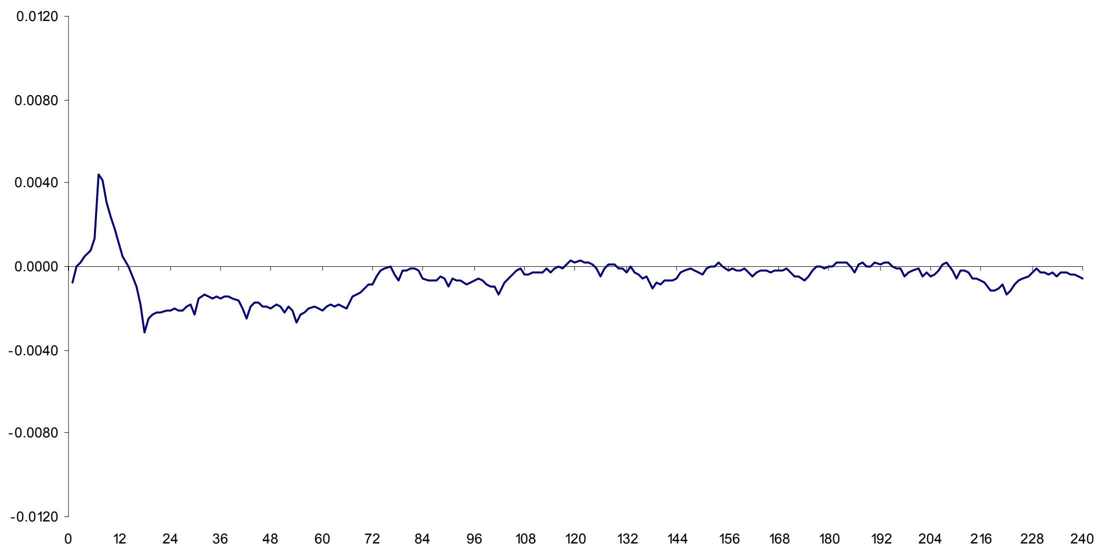
Table 8
Controlling for Earnings Announcements

Every month stocks are grouped into ten portfolios (with equal number of stocks in each portfolio) according to various categories based on past performance. For example, the trading strategy that is formed based on past annual returns during years 2 through 5 ranks stocks according to their average returns during the historical lags 24, 36, 48, and 60. The stocks in each portfolio are assigned equal weight, and the portfolios are rebalanced monthly. The sample includes only NYSE/AMEX-listed firms whose earnings announcements are recorded on COMPUSTAT. The table below reports the portfolio returns using (a) the entire sample, (b) only returns during months of earnings announcements, and (c) only non-earnings-announcement months. The average monthly returns of the various trading strategies for the period January 1965 through December 2002 (456 months) are reported below, as well as the corresponding *t*-statistics (two digit numbers).

Strategy		All	Earnings	non-Earnings
Year 1	All	0.0142	0.0219	0.0122
		5.02	6.15	4.20
	Annual	0.0127	0.0106	0.0127
		7.49	4.25	7.13
	Non-Annual	0.0110	0.0191	0.0091
		3.60	5.34	2.91
Years 2-5	All	-0.0101	-0.0124	-0.0097
		-4.23	-4.27	-3.84
	Annual	0.0076	0.0078	0.0067
		5.56	3.30	4.67
	Non-Annual	-0.0120	-0.0138	-0.0120
		-4.81	-4.83	-4.53
Years 6-10	All	-0.0041	-0.0043	-0.0042
		-3.18	-1.91	-2.84
	Annual	0.0066	0.0074	0.0062
		5.28	3.20	4.56
	Non-Annual	-0.0056	-0.0057	-0.0055
		-4.26	-2.55	-3.59
Years 11-15	All	0.0001	-0.0013	0.0005
		0.11	-0.57	0.40
	Annual	0.0061	0.0085	0.0054
		5.25	3.92	3.95
	Non-Annual	-0.0018	-0.0026	-0.0018
		-1.48	-1.18	-1.34
Years 16-20	All	-0.0020	0.0031	-0.0028
		-1.65	1.22	-2.03
	Annual	0.0062	0.0058	0.0045
		4.88	2.08	3.09
	Non-Annual	-0.0028	-0.0005	-0.0037
		-2.19	-0.18	-2.58

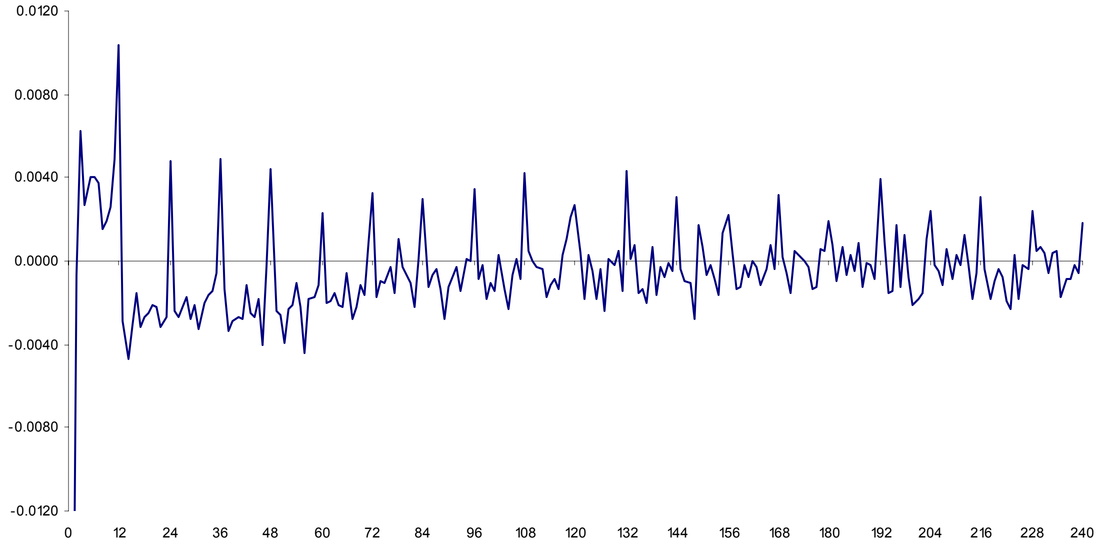


Panel A: Average returns as a function of historical lag

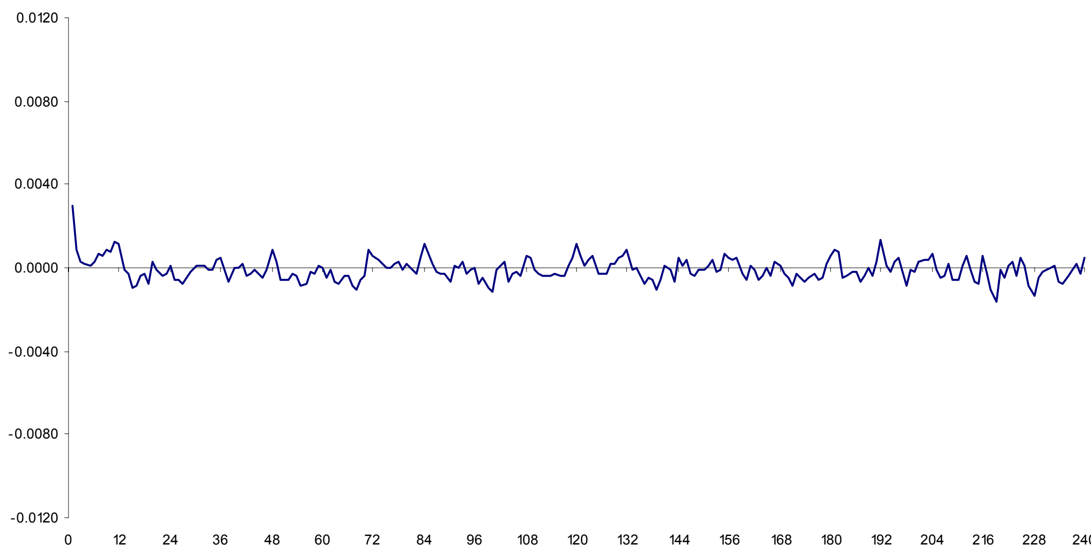


Panel B: Smoothed graph

Fig. 1. Returns of Winners-minus-Losers Strategies. Every month stocks are grouped into ten portfolios (with equal number of stocks in each portfolio) according to the performance during one month in the past. The stocks in each portfolio are assigned equal weight, and the portfolios are rebalanced monthly. The average monthly return difference between the highest past-performing decile and the lowest past-performing decile is then calculated for the period January 1965 through December 2002 (456 months). Panel A plots the average monthly returns for 240 different trading strategies, each corresponding to one historical month out of the past 20 years. Panel B presents a smoothed version of Panel A by replacing each observation in Panel A with the average return using a moving window of [–5,+5]. The analysis uses NYSE/AMEX-listed stocks.



Panel A: Intra-Industry



Panel B: Inter-Industry

Fig. 2. Industry-Adjusted Returns of Winners-minus-Losers Strategies. Every month stocks are grouped into ten portfolios (with equal number of stocks in each portfolio) according to the performance during one month in the past. The stocks in each portfolio are assigned equal weight, and the portfolios are rebalanced monthly. The monthly return of every stock is then decomposed into intra- and inter-industry components. The intra-industry component is the monthly return of the stock excess of its industry group return, and the inter-industry component is the monthly return of the industry group itself. The industry groups (twenty groups) are formed according to the classification in Moskowitz and Grinblatt (1999). The average monthly return difference (for both intra- and inter-industry components) between the highest past-performing decile and the lowest past-performing decile is then calculated for the period January 1965 through December 2002 (456 months). The figure plots the average monthly returns for 240 different trading strategies, each corresponding to one historical month out of the past 20 years. The analysis uses NYSE/AMEX-listed stocks.